CAMBRIDGE O LEVEL COMPUTER SCIENCE (2210 / 0478)

Paper 1: Computer Systems | Topic-Wise Question Bank (2024 – 2026 Series)

TOPIC 02: DATA TRANSMISSION

Exam Sources: Cambridge 2210 / 0478 (May/June 2023, Oct/Nov 2023, May/June 2024)

Question 1 [2210/12/M/J/24 Q2 (2024 Series)] [Total: 6 marks]

(a) Describe the differences between serial data transmission and parallel data transmission. **[2 marks]**

(b) Explain what is meant by simplex, half-duplex, and full-duplex data transmission, providing one example for each. **[3 marks]**

(c) State one benefit of using a USB (Universal Serial Bus) connection. **[1 mark]**

✓ Marking Scheme / Model Answer:

(a) Serial: Bits sent one at a time sequentially over a single wire [1 mark]. Parallel: Multiple bits sent simultaneously across multiple parallel wires [1 mark].
(b) Simplex: One direction only (keyboard -> computer) [1 mark]. Half-duplex: Both directions, but one at a time (walkie-talkie) [1 mark]. Full-duplex: Both directions simultaneously (phone call / broadband) [1 mark].
(c) Automatically detected / high speed / backwards compatible / plug and play / carries power [1 mark].

Question 2 [2210/12/M/J/23 Q2 (2023 Series)] [Total: 6 marks]

(a) Data transmitted across a network is split into packets. Identify the three main components of a data packet. **[3 marks]**

(b) State two items of information found in a packet header. **[2 marks]**

(c) State the role of the packet trailer. **[1 mark]**

✓ Marking Scheme / Model Answer:

(a) 1. Packet Header [1 mark] 2. Payload (the actual data) [1 mark] 3. Packet Trailer [1 mark]
(b) Packet Header items: [Any 2]
- IP address of sender
- IP address of destination / receiver
- Sequence number of packet (to reassemble in order)
- Packet size

(c) Packet Trailer: Contains error detection code (e.g. checksum / parity bit) to check if packet arrived without corruption [1 mark].

Question 3 [2210/11/O/N/23 Q3 (2023 Series)] [Total: 5 marks]

(a) Explain how Automatic Repeat Request (ARQ) works to ensure error-free data transmission. **[3 marks]**

(b) Explain how symmetric encryption differs from asymmetric encryption. **[2 marks]**

✓ Marking Scheme / Model Answer:

(a) ARQ uses positive/negative acknowledgements (ACK / NAK) and timeouts [1 mark].
Sender sends data block and starts a timer [1 mark].
If ACK is received before timeout, next packet is sent; if NAK is received or timeout expires without ACK, packet is retransmitted automatically [1 mark].
(b) Symmetric: Uses one single shared key to both encrypt and decrypt [1 mark].
Asymmetric: Uses a mathematically linked pair of keys — public key (encrypt) and private key (decrypt) [1 mark].